

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (EE) Sem - IV Examination, 2011
EE - 203 Electrical Measurement and Measuring Instruments

Date: 12.05.2011, Thursday Time: 10:00a.m to 1:00p.m Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary.

SECTION I

- Q-1 A. Explain in detail kelvin's double bridge. 3
- B. Define the following terms related to measuring instruments (Any Four): 4
- True Value
 - Accuracy
 - Precision
 - Sensitivity
 - Drift
- C. The resistance of a circuit is found by measuring current flowing and the power fed into the circuit. Find the limiting error in the measurement of resistance when the limiting error in the measurement of power and current are respectively $\pm 1.5\%$ and $\pm 1.0\%$. 3
- Q-2 A. Explain with a neat sketch and phasor diagram Anderson bridge. 5
- B. Explain difficulties encountered in the measurement of high resistances. 3
- C. A sheet of Bakelite 4.5 mm thick is tested at 50 Hz between electrodes 0.12m in diameter. The Schering bridge shown in Fig. 1 employs a standard air capacitor C_2 of 106 pF capacitance, a non-reactive resistance R_4 of $1000/\pi \Omega$ in parallel with a variable capacitor C_4 , and a non-reactive variable resistance R_3 . Balance is obtained with $C_4 = 0.5\mu F$ and $R_3 = 260\Omega$. Calculate capacitance, power factor and relative permittivity of sheet. 4

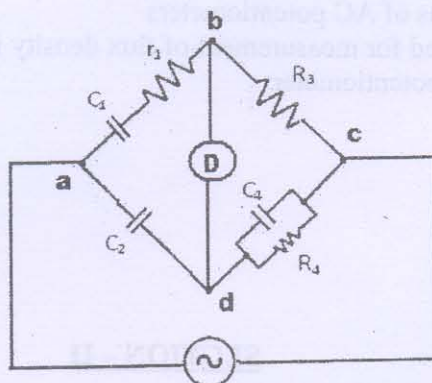


Fig. 1

OR

- Q-2 A. Explain with a neat sketch and phasor diagram De Sauty's bridge. What are the limitations of this bridge? Also suggest necessary modifications. 5

- B. A bridge consists of following: 4

Arm ab:- a choke coil having a resistance R_1 and L_1

Arm bc:- A non-inductive resistance R_3

Arm cd:- capacitance C_4 in series with resistance R_4

Arm da:- a non-inductive resistance R_2

When this bridge shown in Fig. 2 is fed from a source of 500-Hz, balance is obtained under following conditions: $R_2 = 2410\Omega$; $R_3 = 750\Omega$; $C_4 = 0.35\mu F$; $R_4 = 64.5\Omega$.

The series resistance of capacitor is 0.4Ω . Calculate resistance and inductance of the choke coil. The supply is connected between a and c while the detector is connected between b and d.

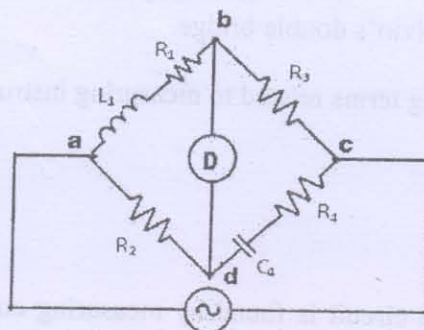


Fig. 2

- C. Explain Ammeter-Voltmeter method for measurement of medium resistances 3

- Q-3 A. Explain in detail Drysdale Polar potentiometer 5

- B. Explain applications of DC potentiometers 2

- C. Explain different methods for determination of B-H curve of a specimen 6

OR

- Q-3 A. Explain in detail Gall-Tinsley A.C. potentiometer 5

- B. Explain applications of AC potentiometers 2

- C. Discuss method used for measurement of flux density in ring specimen 3

- D. Explain Magnetic potentiometer 3

SECTION - II

- Q-4 A. Describe the constructional details of an attraction type Moving Iron instrument with the help of a neat diagram. 6

- B. Explain briefly Electrodynamometer type synchroscope 4

- Q-5 A. A potential transformer with ratio 1000/100 V, has following parameters: 7
- Primary resistance = 94.5Ω
 - Secondary resistance = 0.86Ω
 - Primary reactance = 66.2Ω
 - Total equivalent reactance = 110Ω
 - No load current = 0.02 A at 0.4 power factor

Calculate:-

- Phase angle error at no load
 - Burden in VA at unity power factor at which the phase angle will be zero
- B. Explain Mercury Loop test 5

OR

- Q-5 A. Draw the phasor diagram of C.T. and derive the equation of transformation ratio and phase angle. 7
- B. Explain with a neat figure Varley loop test 5

- Q-6 A. Describe with a neat diagram the constructional details of PMMC instrument 7
- B. Explain Weston type frequency meter 6

OR

- Q-6 A. Explain factors affecting performance of rectifier type instruments 7
- B. Explain Phase sequence indicator 6